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Modelling hydrological processes and aluminium leaching in an acid soil at Llyn Brianne, Mid-Wales

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ABSTRACT

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Recent research has shown that enhanced acid deposition on to the canopies of commercial conifer forests in upland Wales has contributed to the release of Al into soil and stream waters. Relatively little is known about how soil hydrological and chemical processes interact in response to acid deposition and afforestation to regulate the transfer of Al-rich soil water into streams. A field-based hydrochemical study monitored the hydrology and soilwater chemistry of an afforested stagnopodzol at Llyn Brianne (Mid-Wales) during a hydrological year. Water moved vertically through the soil profile, aided by root-promoted macropores. Soil water fluxes, estimated by the LEACHM model, indicated that 89% of effective precipitation drained vertically through the soil profile. Precipitation reaching the soil was acidified and enriched in anions during its passage through the forest canopy. Further acidification occurred as water moved into the soil and Al was mobilized, probably by cation exchange processes. Leaching losses of Al from the soil profile occurred during all major hydrological events and the total flux was estimated at $3.39 \text{ kmol}_e \text{ ha}^{-1} \text{ year}^{-1}$. It is suggested that Al-rich soil water draining from the base of the stagnopodzol profile can provide a significant input to streams during acid episodes.

INTRODUCTION

Atmospheric deposition of acidic oxide pollutants has been implicated in the acidification of soils and stream water in parts of upland Wales (United Kingdom Acid Waters Review Group, 1989). The afforestation of catchments dominated by acidic soils exacerbates acidification by increasing dry and occult deposition of pollutants on to forest canopies (Fowler et al., 1989). Consequently, streams draining afforested catchments in upland Wales can be

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acidic and Al-bearing, particularly during periods of high discharge (Reynolds et al., 1986). High levels of Al in stream water have been shown to be toxic to aquatic organisms and have resulted in a reduced diversity of species in forest streams (Ormerod et al., 1989).

The release of Al into stream water is controlled by two main factors: the chemical interactions governing the mobilization of Al into soil water, and the soil hydrological pathways which transfer Al-rich soil water into streams (Bache, 1984). Although soil chemical processes have been intensively studied in recent acidification investigations, the detailed examination of hydrological pathways has usually been neglected (Neal et al., 1989). A corollary is that it is rarely possible to identify the hydrological processes involved in the generation of acid runoff in a given catchment (Bishop et al., 1990). Moreover, it is usually impossible to quantify the fluxes of water and solutes through and from acid soils (Neal et al., 1990). Identifying hydrological pathways requires intensive field monitoring. In addition, quantifying soil water and solute fluxes requires the simulation of soil hydrology by mathematical modelling. This paper presents the results of one such study carried out at Llyn Brianne in Mid-Wales. The investigation demonstrates the importance of considering soil hydrological pathways when examining Al mobilization; a knowledge of soil water and solute fluxes is a prerequisite to understanding the impact of acid deposition and land-use change on stream hydrochemistry.

STUDY AREA AND METHODS

Experimental setting

A major catchment study was established at Llyn Brianne, Mid-Wales, in 1984 to investigate the effects of acid deposition, afforestation and land management on streamwater chemistry and aquatic ecology (Gee, 1990). The area experiences acid deposition: rainfall has a mean pH of 4.9 and wet deposition of H^+ and S are 0.8 and 15 kg ha⁻¹ year⁻¹, respectively (Donald et al., 1990). Extensive afforestation, mainly with Sitka spruce (*Picea sitchensis*), at Llyn Brianne during the 1950s and 1960s has increased the dry and occult deposition of acidic oxides on to forest canopies; acidic, Al-bearing stream water and depleted aquatic ecosystems in afforested catchments have resulted (Stoner et al., 1984). During storm episodes, forest streams have a low pH (less than 4.5) and high concentrations (greater than 25 $\mu\text{mol dm}^{-3}$) of toxic inorganic Al species such as Al^{3+} (Goenaga and Williams, 1988). Despite a decade of research at Llyn Brianne, little is known about the hydrological processes governing the transfer of Al-rich soil water into streams.

Field study

Stagnopodzols are an important soil type in acidic catchments in upland Wales; they cover approximately 20% of the afforested land at Llyn Brianne (Pyatt, 1977), and have been implicated as a probable source of acid, Al-bearing storm runoff (Hornung et al., 1987). Stagnopodzols at Llyn Brianne have developed in coarse periglacial drifts derived from the underlying Lower Palaeozoic bedrock. Pedogenic weathering of this material has produced a 10–15 cm thick silt–clay E_g horizon. This has impeded drainage, causing surface water gleying and the accumulation of a 25–30 cm deep O horizon of well-humified peat. At 35–45 cm a thin, discontinuous ironpan is found above an ochreous, gravelly B_s horizon. At 60 cm a little-altered C horizon occurs, which merges into the underlying slope drift at 80 cm. The bedrock is usually within 1 m of the soil surface. The soils have been strongly influenced by recent afforestation. A 5–10 cm thick L horizon of spruce needles covers the soil surface and abundant fine tree roots are concentrated in this and the underlying O horizon. Coarse tap roots penetrate both the E horizon and the ironpan, and rooting is common to a depth of 70 cm. The soils have been ploughed parallel to the slope; however, the plough furrows are shallow (10–20 cm) and have not penetrated the mineral subsoil. Ditches aligned to the slope contour have been dug at regular intervals to collect drainage water.

A representative plot on a 5° hillslope was established to monitor the hydrology and soil water chemistry of a stagnopodzol which was afforested with Sitka spruce in 1961 (Fig. 1). The site, located in the afforested control catchment of the Llyn Brianne Project, was monitored from 1 October 1988 to 30 September 1989. Precipitation was measured by a recording raingauge close to the hillslope and an adjacent forest plot was instrumented with seven throughfall gauges and six stemflow collectors. Throughfall and stemflow were measured fortnightly and samples taken for analysis. Soil water pressure potentials (p) were measured along a hillslope transect with a distributed network of tensiometers. Three sets of tensiometers were located in the lower (A), middle (B) and upper slope (C) areas respectively. Replicate sets were placed in a plough furrow (F) and ridge (R). Tensiometers in each set were located at depths of 25, 35, 55 and 75 cm to correspond to the O, E, B and C horizons respectively. Pressure potentials were recorded twice-weekly during the study year and hourly during storm episodes. A piezometer in the lower slope was used to monitor the depth to the water table.

Lysimeters were used to sample soil water at fortnightly intervals. Porous cup suction samplers were located in the O, E, B and C horizons (three replicates in each horizon). The ceramic cups were leached in the laboratory prior to installation and allowed to equilibrate in situ for 3 months prior to

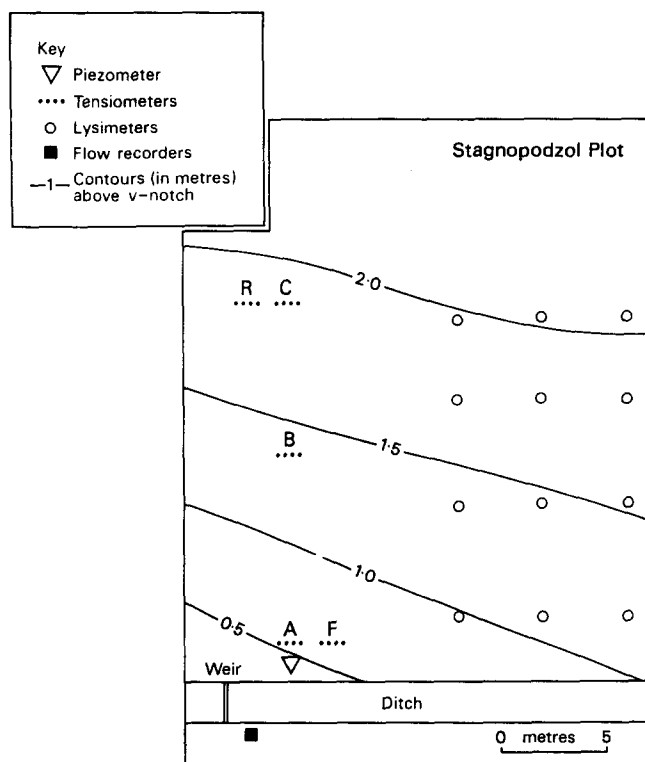


Fig. 1. Instrumented stagnopodzol study plot.

sampling; these procedures have been shown to avoid any major sample contamination (Hughes and Reynolds, 1988, 1990). Three plastic tensionless tray lysimeters were installed beneath the L horizon to collect litter leachates.

Analytical techniques

All solutions were collected in clean polyethylene bottles and were filtered ($0.45\ \mu\text{m}$ membrane) on return to the laboratory; analysis followed the day after collection. Inductively coupled plasma emission spectroscopy was used to determine Al, Ca, Mg, Na, and K. Dissolved organic carbon (DOC), SO_4 , Cl, NO_3 , and SiO_2 were determined colorimetrically using auto-analyzer techniques. Aluminium was assumed to be monomeric (Al_m). This was operationally fractionated into non-labile Al_m (mainly organically complexed; Al_o) and labile Al_m (mainly inorganic; Al_i) using the exchange column procedure described by Driscoll (1984). Unfiltered subsamples were used for electro-metric pH measurements.

The physical and chemical characteristics of each soil horizon were

TABLE 1

Physical properties of the stagnopodzol soil horizons

Horizon	Sand (%)	Silt (%)	Clay (%)	Bulk density (g cm ⁻³)	LOI (%)
O	n.a.	n.a.	n.a.	0.654	79.7
E	32.1	38.8	29.0	1.120	9.8
B	64.1	22.3	13.5	1.386	5.9
C	61.4	31.2	7.4	1.588	4.1

n.a. = not available.

determined on three replicate samples. Texture, bulk density and loss-on-ignition (LOI) were measured using standard methods (Avery and Bascomb, 1974). The soil moisture characteristic curve for each horizon was derived using pressure plate apparatus to determine the volumetric moisture content (θ) of undisturbed cores (100 cm³) at pressure potentials of -1, -2, -3, -5, -33, -100, -300 and -500 kPa. The saturated hydraulic conductivity, $K(s)$, of the soil was measured in the field using a ring permeameter on five replicate cores for each horizon (Talsma, 1969). Unsaturated hydraulic conductivity, $K(\theta)$, was estimated using the method described by Campbell (1974). Soil chemical properties were measured using standard methods described by Allen (1989). Briefly, soil pH was measured in 1:2.5 w/w water suspension. Exchangeable base cations were determined by atomic absorption spectrophotometry following extraction with 1M ammonium acetate at pH 7. Exchangeable Al was also determined by AAS following extraction with 1M KCl. The same KCl extracts were used to determine exchangeable H⁺ by Gran Titration. Cation exchange capacity (CEC) was estimated from the sum of exchangeable base cations, Al and H⁺.

RESULTS AND DISCUSSION

Soil hydrology

The hydraulic characteristics of the soil profile exhibited marked vertical stratification (Table 1). The well-structured O horizon had the highest saturated hydraulic conductivity; permeability decreased in the denser, silt-clay E horizon and then increased in the coarse textured subsoil (Table 2). All horizons were characterized by marked spatial variability in $K(s)$; this may be associated with the location of macropores that were often related to tree root penetration. The soil moisture characteristic curve for each horizon reflects

TABLE 2

Saturated hydraulic conductivity (cm h^{-1}) and unsaturated hydraulic conductivity at -3 kPa (geometric mean of five samples) in stagnopodzol horizons

Horizon	Mean $K(s)$	Minimum $K(s)$	Maximum $K(s)$	$K(\Theta)$ at -3 kPa
O	83.4	8.3	166.7	0.003
E	4.3	2.3	8.0	0.001
B	7.9	0.9	20.8	0.024
C	4.7	1.9	19.0	0.010

these hydraulic properties (Fig. 2). The O and E horizons were more retentive than the more freely draining lower profile (Fig. 2). However, all horizons lost water most rapidly between suctions of 0 and 10 kPa as the primary and secondary structural pores drained. Correspondingly, the hydraulic conductivity of each horizon decreased markedly as suctions increased (Table 2).

During the study year, the soil water regimes at each of the five tensiometer sets were generally similar to that of set B shown in Fig. 3. Some small but significant differences are discussed by Soulsby (1992). Rainfall during this period totalled 1909 mm, 40% of which fell between January and March 1989 (days 93–182). Interception losses from the forest canopy accounted for 335 mm of incident precipitation, whilst 1391 mm and 183 mm reached the forest floor as throughfall and stemflow respectively. The drought of 1989 resulted in summer rainfall being less than half the average at Llyn Brianne.

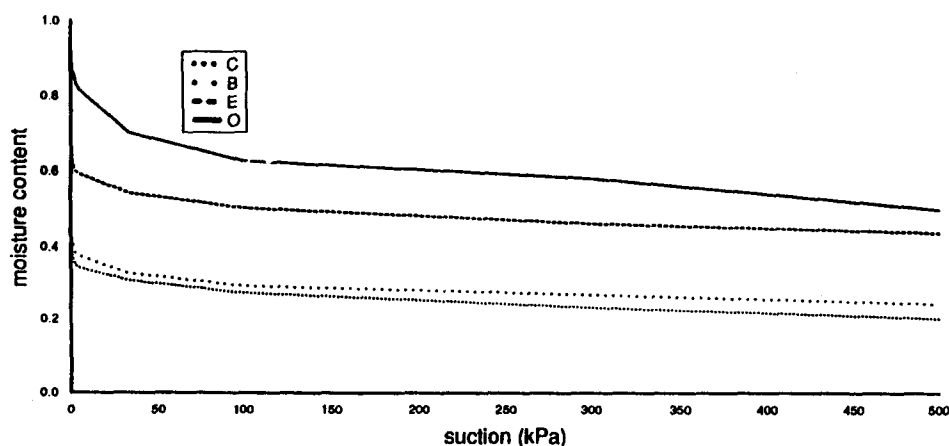


Fig. 2. Stagnopodzol soil moisture characteristic curve showing volumetric moisture content changes with increased suction for each horizon.

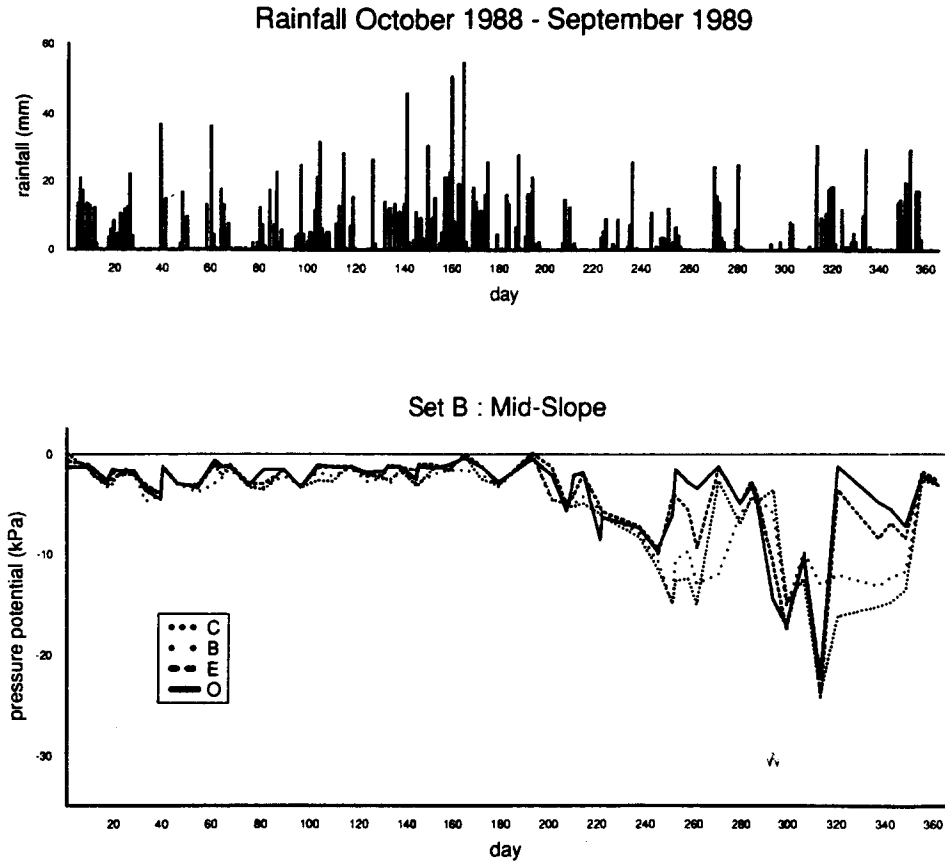


Fig. 3. Rainfall and soil hydrological regime (October 1988 to September 1989) at set B on the stagnopodzol study plot.

Consequently, a marked seasonal dichotomy in soil hydrological conditions was observed.

During the winter, all horizons were wet, with pressure potentials remaining above -6 kPa between October and April (days 1–182). Potentials were slightly higher in the more retentive O and E horizons, though all the horizons wetted significantly during winter storm episodes. The hillslope remained largely unsaturated throughout the year, though positive pressures were detected in the B and C horizons of the lower slope tensiometers (set A) during the wet March period. Detailed examination of storm hydrological pathways showed that the low slope angle and the abundance of root-promoted macropores (penetrating the E horizon and ironpan) dictated that the profile was dominated by vertical drainage (Soulsby, 1992). Lateral water

movement was only detected at a depth of 1 m, immediately above the impermeable shale bedrock (Soulsby, 1991).

The drying of the soil profile during the summer period commenced in May. Initially, drying was most marked in the more freely draining B and C horizons, though eventually the O and E horizons responded to the transpiration demands of tree roots. Rainfall during the summer mainly replenished moisture deficits in the surface horizons, though heavy rainfall in September returned the whole profile to pre-summer conditions by the end of the study year.

The dominant vertical hydrological pathways observed are similar to those observed in hydrological studies at other afforested stagnopodzol sites in the UK (Stevens et al., 1989; Wheater et al., 1990). The main control on the soil hydrology appears to be the ability of tree roots to penetrate the eluvial horizon and ironpan which impede drainage at some sites (Pyatt and Smith, 1983).

Modelling soil water fluxes

The tensiometer data provided the basis for calibrating a hydrological simulation model that enabled soil water fluxes to be calculated. The leaching and chemistry estimation model (LEACHM) contains a finite-difference model of soil water flow developed and described by Wagenet and Hutson (1989). The flow model simulates vertical water movement in the unsaturated soil matrix and is based on numerical solutions of the Richards equation:

$$\frac{\partial \Theta}{\partial t} = \frac{\partial}{\partial x} [K(\theta) \cdot \partial h / \partial x] - U(x, t) \quad (1)$$

where θ is the volumetric water content ($\text{m}^3 \text{m}^{-3}$), h is the hydraulic head (mm), $K(\theta)$ is the hydraulic conductivity (mm day^{-1}) at water content θ , t is the time (day), x is the depth (mm) and U is a sink term representing water loss per unit time (day) by transpiration.

The model was parameterized to simulate the upper 80 cm of the stagnopodzol profile. Measured values of $K(s)$, bulk density and the retentivity characteristics of each horizon were used in the parameterization. The model calculates $K(\theta)$ from these parameters using Campbell's retentivity equation (Hutson and Wagenet, 1991). The surface boundary fluxes used in the model were the measured daily totals of effective rainfall and weekly transpiration losses. Transpiration losses were estimated from published water balance studies in Welsh Sitka spruce stands (Hudson, 1988), with weekly totals ranging between 2 mm in winter and 20 mm in summer. The transpiration losses were allocated according to a specified rooting pattern, which in this

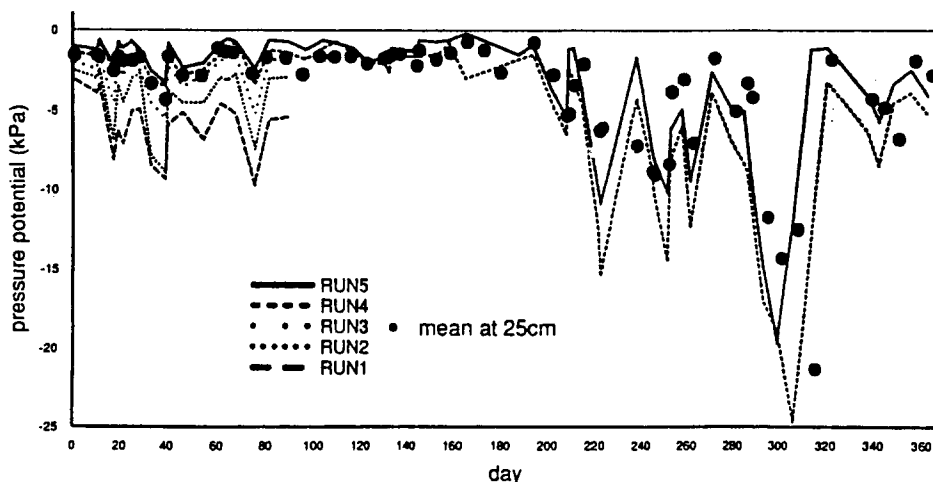


Fig. 4. LEACHM model calibration for the stagnopodzol O horizon.

case concentrated most losses in the upper 30 cm of the profile. The lower soil boundary condition was specified as being freely draining.

The model was used to predict soil water pressure potentials and water fluxes at 5 cm depth intervals for daily time steps. The model was calibrated over the first three months for which tensiometer data were available (October–December 1988) and validated against the tensiometer data for the remainder of the year; thus some degree of independence was retained between the calibration and validation data sets. The model calibration for the soil O horizon is shown in Fig. 4. The mean pressure potential of the O horizon at tensiometer sets A, B and C was used for the calibration. As the model sought to simulate ‘average’ conditions over the experimental plot, it was assumed that this was the most appropriate way of incorporating the spatial variability in soil moisture conditions. As the soil hydrology at sets F and R was not markedly different from sets A and C respectively, the effect of the shallow ploughing was ignored. Measured pressure potentials were used to initialize the model; however, in run 1 it became apparent that predicted pressure potentials were being underestimated, perhaps because the measurements of $K(s)$ were overestimates (Hutson and Wagenet, 1991). This is a common problem in modelling studies as the spatial and temporal variation of $K(s)$ in soils is so large that it is difficult to characterize accurately (Biggar and Nielson, 1976). Consequently, values of $K(s)$ were progressively reduced for each soil horizon until simulated pressure potentials fitted those measured during the calibration period (Van Grinsven et al., 1987). Optimized values of $K(s)$ were 5 to 90 times lower than those measured. The discrepancies could

be partly explained by the model not considering preferential flow in soil macropores, which are important hydrological pathways in the upper profile. By run 4 the model was validated against the data set for the whole study year. Conductivity estimates were still too high; despite good convergence of simulated and measured pressure potentials during the winter, the simulation predicted greater summer drying than actually occurred. Further optimization of $K(s)$ gave a more representative simulation for the whole period in run 5. The same procedure was repeated for the E, B and C horizons until simulated pressure potentials fitted those measured.

The final simulation of pressure potentials is plotted against those measured in the O and B horizon at the three tensiometer sets in Fig. 5 (fits of similar quality were obtained for the E and C horizons). The simulations show good agreement with the observations over the winter period but diverge during the summer drying when spatial variability in soil moisture conditions increased. Nevertheless, the simulation in all horizons generally remained within the range of observed values. The resultant weekly water fluxes simulated for the soil profile are shown in Fig. 6. Substantial vertical fluxes occur in response to all major rainfall events. The losses of water at the base of the C horizon peaked at over 80 mm in the wettest week during March around day 160. The very low fluxes during the summer drought are also evident. The summary of the profile water budget shows that 89.3% of effective precipitation drained from the C horizon (Table 3). The summer drying resulted in a small water deficit of 28 mm at the end of the study period.

It is difficult to evaluate the accuracy of these flux estimates without further detailed investigation and error analysis beyond the scope of this study. The convergence of measured and simulated soil moisture conditions is at least equal to the levels of agreement that have been accepted as model validation in similar modelling studies (Hauhs, 1986; Van Grinsven et al., 1987). Moreover, it is encouraging that the simulation was most accurate during the winter period when water fluxes are greatest. However, it is important to recognize the limitations of the model outputs. A major feature of the stagnopodzol hydrology, for example, is the importance of preferential flow paths through the impeding E horizon and ironpan. The movement of water through these horizons will therefore be non-uniform (Bevan and Germann, 1982). By predicting an average flux in the soil matrix, the model does not account for such spatial variability and macropore flow. Nevertheless, it is still likely that the flux estimates give a reasonable indication of the hydrology of the stagnopodzol over the study plot as a whole.

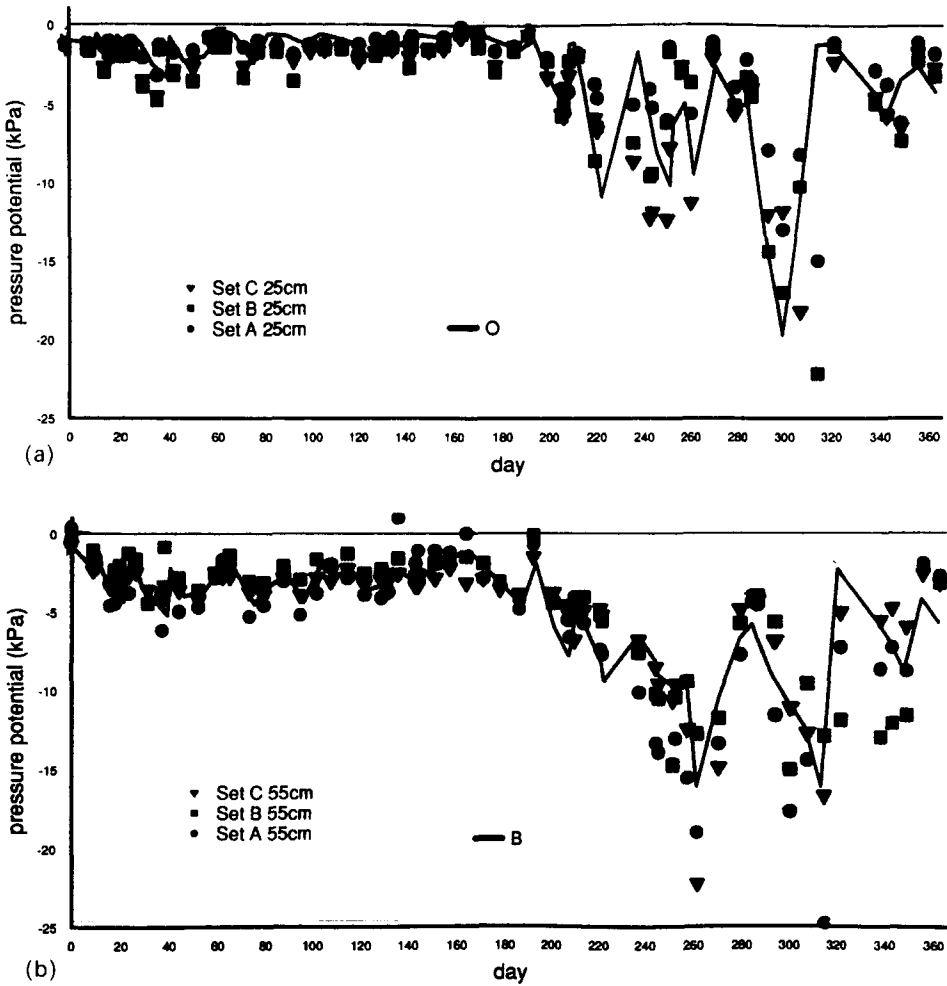


Fig. 5. Final simulated (line) and measured (points) pressure potentials in (a) stagnopodzol O and (b) stagnopodzol B horizons.

Soil water chemistry

The stagnopodzol is strongly acidic and the chemical composition of soil drainage waters reflected changes in the soil chemistry. Soil pH was lowest in the organic-rich O horizon and increased in the mineral subsoil (Table 4). Base saturation was low throughout the profile owing to the base-poor status of Lower Palaeozoic rocks and the intensively leached nature of the soil (Adams, 1974); the soil cation exchange complex was therefore dominated by Al.

Although the precipitation at Llyn Brianne was acidic, it was further acidified and enriched in atmospherically derived anions in its passage

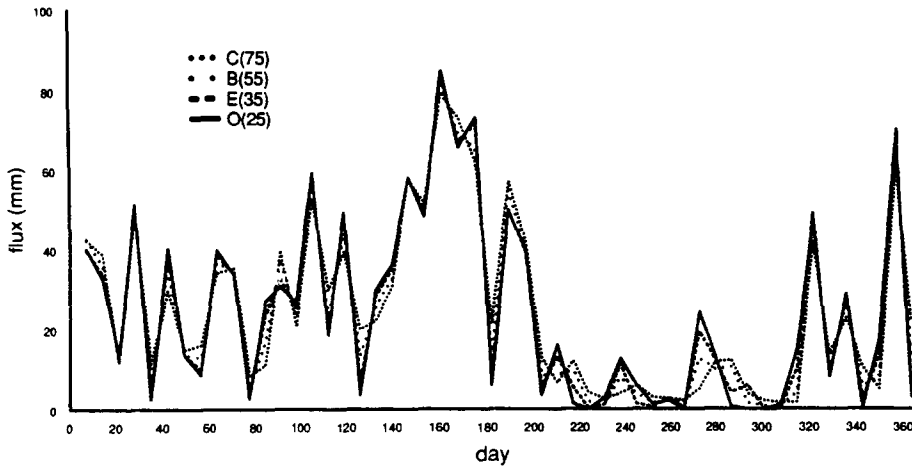


Fig. 6. Simulated soil water fluxes (October 1988 to September 1989) in stagnopodzol at depths of 25, 35, 55 and 75 cm.

through the forest canopy (Table 5). This was primarily caused by the washing out of material accumulated from dry and occult deposition (Hornung et al., 1990a). Water was further acidified as it moved through the soil profile (Table 6). Acidity was highest in the surface horizons, with H^+ ions being consumed in the lower mineral horizons. Aluminium was mobilized in the L and O horizons, though the largest increase in concentration occurred in the mineral subsoil. Inorganic Al (Al_i) was the dominant fraction of Al_m , with Al_o concentrations remaining low but constant throughout the profile. Thermodynamic analysis revealed that Al^{3+} was the dominant inorganic species, accounting for up to 80% of Al_i in each horizon (Soulsby, 1991). The ionic composition of soil water was dominated by solutes of pollutant and marine origin. Sulphate and Cl accounted for over 90% of the anionic charge, whereas Na was the dominant cation. Concentrations of other base cations were low. Levels of DOC were highest in the organic surface horizons, whereas SiO_2 concentrations were highest in the intensively weathered E horizon.

TABLE 3

Modelled soil hydrological budget (1 October 1988 to 30 September 1989) for stagnopodzol

Precipitation input to soil	1575 mm
Transpiration losses from soil	196 mm
Drainage losses from C horizon	1407 mm
Change in profile water content	- 28 mm

TABLE 4

Exchangeable cations in bulked samples from stagnopodzol

Horizon	pH	Na	K	Ca	Mg	H	Al	CEC
O	3.46	7.0	4.4	3.5	3.3	1.8	115.4	135.4
E	3.64	3.0	0.7	1.8	1.6	0.5	95.8	103.4
B	3.80	3.1	0.9	1.9	1.4	0.4	46.9	54.6
C	3.92	2.1	0.7	1.5	0.5	0.4	40.2	45.4

All values except pH in $\text{mmol}_c \text{kg}^{-1}$.*Aluminium mobilization into soil waters*

Although Al in acid soils in upland Wales is ultimately derived from the weathering of aluminosilicate minerals, recent investigations have shown that it is extremely unlikely that mineral solubility controls regulate Al_i levels in soil water (Neal et al., 1990). Thermodynamic considerations show that soil water is undersaturated with respect to Al-bearing minerals such as gibbsite in this and similar soils (Soulsby, 1991). It has therefore been suggested that cation exchange reactions may control the short-term release of Al_i into soil water (Neal et al., 1989). A regression analysis revealed very strong correlations between Al_i and total anion concentrations in the stagnopodzol soil water (Table 7). Up to 80% of the variation in Al_i could be explained by variations in the total anionic charge. These correlations are consistent with the mobile anion hypothesis, which proposes that the anionic strength of soil water determines its ability to mobilize Al and other cations from the cation exchange complex (Seip, 1980).

Theoretically, cation exchange reactions can be simply described by a non-specific thermodynamic exchange equation involving a solid phase of

TABLE 5

Weighted mean concentration and range ($\mu\text{mol}_c \text{dm}^{-3}$) of selected solutes in bulk precipitation, throughfall and stemflow at Llyn Brianne (October 1988 to September 1989)

	Precipitation	Throughfall	Stemflow
pH	4.98	4.37	4.05
H^+	11(1–100)	43(10–200)	89(16–316)
Al_m	3(1–19)	6(3–30)	6(3–25)
Cl	89(14–282)	344(79–1466)	413(76–2403)
SO_4	53(17–215)	181(61–789)	248(33–2207)

TABLE 6

Arithmetic mean concentrations ($\mu\text{mol c dm}^{-3}$) of major solutes ($\pm$ standard error) in soil solution extracted from each stagnopodzol horizon between October 1988 and September 1989

	L	O	E	B	C
pH	3.82	3.67	4.00	4.04	3.97
H ⁺	153 $\pm$ 18	213 $\pm$ 12	101 $\pm$ 5	91 $\pm$ 6	107 $\pm$ 9
Al _i	20 $\pm$ 5	91 $\pm$ 10	277 $\pm$ 15	317 $\pm$ 29	268 $\pm$ 20
Al ₀	4 $\pm$ 1	8 $\pm$ 1	8 $\pm$ 2	9 $\pm$ 2	8 $\pm$ 3
($\mu\text{mol l}^{-1}$)					
Cl	334 $\pm$ 34	498 $\pm$ 29	423 $\pm$ 16	518 $\pm$ 63	303 $\pm$ 10
SO ₄	240 $\pm$ 34	264 $\pm$ 22	262 $\pm$ 11	280 $\pm$ 21	243 $\pm$ 10
NO ₃	68 $\pm$ 36	9 $\pm$ 5	31 $\pm$ 5	34 $\pm$ 11	55 $\pm$ 7
Na	337 $\pm$ 84	445 $\pm$ 35	320 $\pm$ 15	399 $\pm$ 41	248 $\pm$ 11
K	57 $\pm$ 20	13 $\pm$ 6	19 $\pm$ 8	7 $\pm$ 2	5 $\pm$ 2
Ca	41 $\pm$ 6	20 $\pm$ 3	31 $\pm$ 6	25 $\pm$ 4	19 $\pm$ 2
Mg	100 $\pm$ 14	98 $\pm$ 6	72 $\pm$ 3	69 $\pm$ 4	36 $\pm$ 1
SiO ₂	21 $\pm$ 3	68 $\pm$ 6	107 $\pm$ 6	78 $\pm$ 9	71 $\pm$ 7
($\mu\text{mol l}^{-1}$)					
DOC	21 $\pm$ 3	15 $\pm$ 1	7 $\pm$ 1	10 $\pm$ 1	8 $\pm$ 1
(mg l ⁻¹)					

constant CEC:

$$\text{ads}(M) + mN^{n+} = \text{ads}(N) + nM^{m+} \quad (2)$$

where M and N are cations with respective charges $m+$ and $n+$, and $\text{ads}()$ is the exchangeable fraction of each. Over a short timescale, the number of

TABLE 7

Selected correlation coefficients (significant at $p \leq 0.05$) from the stagnopodzol soil solution chemistry data

y	x	Horizon				
		L	O	E	B	C
Al _i	Cl	0.781 ^c	0.555 ^b	—	—	—
Al _i	SO ₄	—	0.623 ^b	—	—	—
Al _i	NO ₃	0.762 ^c	—	—	—	—
Al _i	ΣAnions	0.787 ^c	0.724 ^c	0.465 ^a	0.851 ^c	0.899 ^c

^a $p \leq 0.05$.

^b $p \leq 0.01$.

^c $p \leq 0.001$.

exchange sites greatly exceeds the number of exchangeable cations passing through in the soil water. Thus, even with fluctuations in the composition of the soil solution, the ratio of adsorbed ions remains virtually constant. It follows that as a first approximation, the ratio of a given cation pair in the soil solution also remains constant (Reuss and Johnson, 1986), so that

$$(M^{m+})^n/(N^{n+})^m = \text{constant} \quad (3)$$

From these assumptions, constants can be calculated for the ratios between Al^{3+} and other cation species, and a theoretical relationship between the concentration of a given cation and the total anionic charge in the soil solution can be derived using the procedures described by Reynolds et al. (1988). To simplify calculations, constants between Al^{3+} (approximated by Al_i) and monovalent (M^+) and divalent (M^{2+}) cation groups (where M^+ is the sum of $\text{H}^+ + \text{Na} + \text{K}$, and M^{2+} is the sum of $\text{Ca} + \text{Mg}$) were calculated from the soil water chemistry data. The predicted concentrations of Al^{3+} and other cations agreed well with those measured in the soil solution (Fig. 7), indicating that cation exchange relationships hold over the range of anion concentrations observed during the study period. This, together with the strong correlations in Table 7, implies that Al_i levels in soil water are strongly influenced by cation exchange processes. The importance of cation exchange in regulating the release of Al_i from acidic soils in upland Wales has also been demonstrated by the field investigations of Reynolds et al. (1988) at Plynlimon and laboratory leaching studies by Adams et al. (1990). However, the exact nature of these processes is not yet fully understood; they have been shown by Neal et al. (1990) to be more complex than the relationships implied by the equations used here.

Aluminium leaching fluxes from the soil profile

Fortnightly water fluxes estimated for each soil horizon were multiplied by the solute concentrations measured for each sampling period to calculate the mass flux of solutes through and from the soil profile (Van Grinsven et al., 1987). In view of the uncertainties about the accuracy of the soil water fluxes, these estimates of solute leaching fluxes must be regarded as preliminary.

The vertical flow paths through the soil result in leaching losses in profile drainage below the C horizon (Table 8). The H^+ fluxes show losses from the O horizon and subsequent neutralization with depth. The immobilization of H^+ was wholly accounted for by increased Al_i fluxes. Aluminium was primarily mobilized in the upper profile, with fluxes from the O and E horizons accounting for over 95% of the total profile loss. The flux of anions down the profile was fairly constant and losses from the C horizon indicate

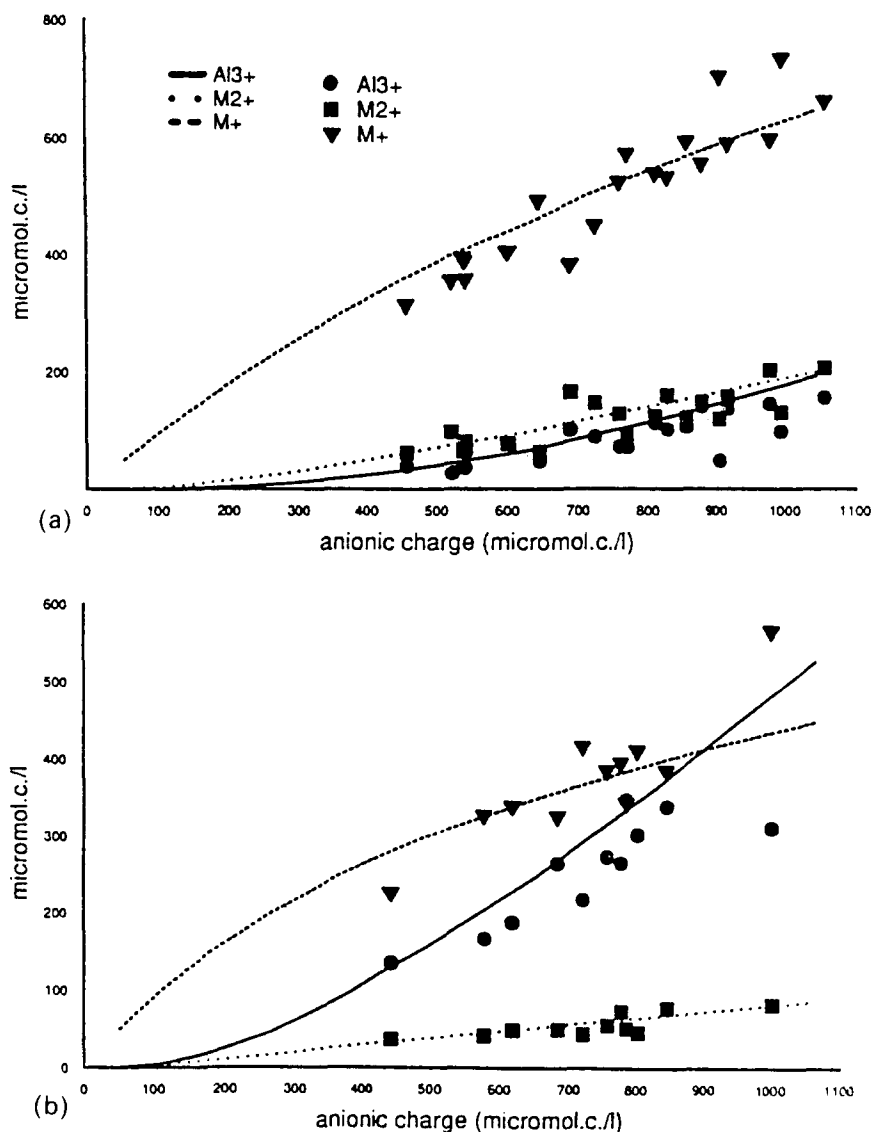


Fig. 7. Cation concentrations in (a) stagnopodzol O and (b) stagnopodzol C horizon soil solutions predicted by cation exchange model (lines) and measured (points).

the ability of mobile anions to leach Al_i from the profile. In each horizon the flux of cations exceeded that of anions; the anion deficit probably reflects the unmeasured loss of organic anions.

The timing and variation with depth of Al_i and H^+ fluxes were plotted in three-dimensions (Fig. 8). These were primarily driven by high water fluxes

TABLE 8

Leaching fluxes ($\text{kmol}_e \text{ ha}^{-1} \text{ year}^{-1}$) of hydrogen, aluminium, base cations (ΣBC), total cations (ΣC) and total anions (ΣA) from each stagnopodzol horizon

	H	Al _i	ΣBC	ΣC	ΣA
O	2.70	0.95	6.56	10.2	9.27
E	1.28	3.15	5.27	9.70	9.70
B	1.06	3.36	5.37	9.79	9.19
C	1.27	3.39	3.66	8.32	7.70

resulting from major hydrological events, indicating that acid, Al-bearing drainage water from ironpan stagnopodzols may contribute to acid episodes in afforested Welsh catchments. The high fluxes of H^+ and Al_i during the winter period, particularly between January and March, are evident, as are high fluxes during events at the end of the dry summer. Leaching losses from the profile were negligible during the summer when base flows in afforested catchments tend to be neutral and Ca-rich (Soulsby, 1991). The plot also illustrates the neutralization of H^+ with depth in the profile and the corresponding release of Al_i . This relationship may support the findings of Adams et al. (1990) who showed in laboratory experiments that H^+ is likely to be the dominant cation displacing Al from cation exchange sites.

The total Al_i leaching loss from the soil profile is less than 0.2% of the estimated total reserves of exchangeable Al (Table 9). Moreover, calculated weathering rates show that the release of Al from minerals within upland soils on Lower Palaeozoic rocks is probably in excess of the annual leaching losses estimated in this study (Evans and Adams, 1975). Thus, there is little evidence to suggest that acid deposition will cause significant depletion of the pool of exchangeable Al.

WIDER IMPLICATIONS

These results demonstrate that Al_i leaching in acid soil profiles is strongly

TABLE 9

Pool of exchangeable Al ($\text{kmol}_e \text{ ha}^{-1}$) in the upper 80 cm of stagnopodzol profile at Llyn Brianne

Horizon	Exchangeable Al
O	112.3
E	98.3
B	52.5
C	42.4

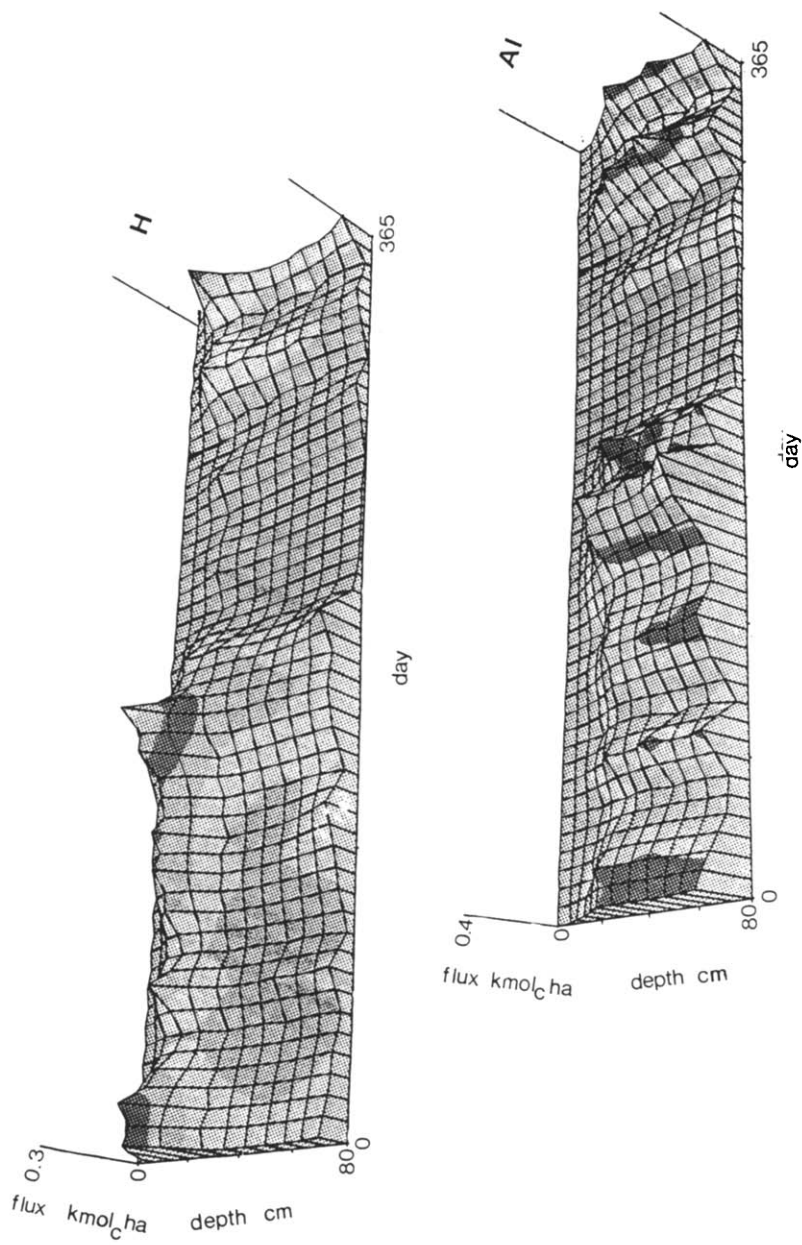


Fig. 8. Weekly leaching fluxes of H⁺ and Al³⁺ in stagnopodzol soil profile between October 1988 and September 1989.

controlled by hydrological factors. Although cation exchange reactions are probably the main chemical process governing Al_i release, Al is transported through and from the soil along the dominant hydrological pathways. Below the soil profile, water is deflected downslope above the impermeable shale bedrock towards drainage channels. Buffering reactions in the drift below the soil are limited and drainage water provides a significant source of acid, Al-bearing storm runoff (Soulsby, 1992). It is likely that similar processes are operative in other afforested stagnopodzols in upland Wales, confirming the views of Reynolds et al. (1986, 1988) that these soils make an important contribution to the catchment hydrochemical response during acid episodes.

Ormerod et al. (1988) have estimated that a 50% reduction of 1984 levels of acid deposition would be needed to prevent further acidification of streams at Llyn Brianne. Despite proposals by the European Community to reduce S emissions by 70%, the enhanced deposition of pollutants on to forest canopies will offset the effects of this directive (Donald and Gee, 1990), and it is likely that SO₄ inputs will contribute to Al leaching from acid forest soils in the foreseeable future. Thus, acid stream water is likely to remain a problem in upland Wales unless afforestation is restricted in acid-sensitive catchments or suitable ameliorative land management strategies (e.g. liming) are implemented (Hornung et al., 1990b).

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REFERENCES

- Adams, W.A., 1974. Pedogenesis of soils derived from Lower Palaeozoic sediments in mid-Wales. Welsh Soils Discussion Group Report, Welsh Soils Discussion Group, Aberystwyth, 15, pp. 61–69.
- Adams, W.A., Ali, A.Y. and Lewis, P.J., 1990. Release of cationic aluminium from acidic soils into drainage water and relationships with land use. *J. Soil Sci.*, 41: 255–268.
- Allen, S.E., 1989. *Chemical Analysis of Ecological Materials*, 2nd edn. Blackwell, Oxford.
- Avery, B.W. and Bascomb, C.L., 1974. *Soil Survey Laboratory Methods*. Soil Survey Technical Monograph No. 6, Harpenden, 43 pp.

- Bache, B.W., 1984. Soil-water interactions. *Philos. Trans. R. Soc. London, Ser. B*, 305: 393-407.
- Bevan, K. and Germann, P., 1982. Macropores and water flow in soils. *Water Resour. Res.*, 18: 1311-1325.
- Biggar, J.W. and Nielson, D.R., 1976. Spatial variability of the leaching characteristics of a field soil. *Water Resour. Res.*, 12: 78-84.
- Bishop, K., Grip, H. and O'Neill, A., 1990. The origins of acid runoff in a hillslope during storm events. *J. Hydrol.*, 116: 35-61.
- Campbell, G., 1974. A simple method for determining unsaturated conductivity from moisture retention data. *Soil Sci.*, 117: 311-314.
- Donald, A.P. and Gee, A.S., 1990. Acid waters in Wales: causes, effects and remedies. In: *Proc. Int. Symp. on Acid Deposition and Water Quality*, Glasgow.
- Donald, A.P., Stoner, J.H., Reynolds, B., Oldfield, F., Rippey, B. and Natakanski, J., 1990. Atmospheric deposition. In: R.W. Edwards, J.H. Stoner and A.S. Gee (Editors), *Acid Waters in Wales*. Kluwer, Dordrecht, pp. 39-53.
- Driscoll, C.T., 1984. A procedure for the fractionation of aqueous aluminium in dilute acidic waters. *Int. J. Anal. Chem.*, 16: 267-283.
- Evans, L.J. and Adams, W.A., 1975. Quantitative pedological studies on soils derived from Silurian mudstones 5. Redistribution and loss of mobilized constituents. *J. Soil Sci.*, 26: 327-335.
- Fowler, D., Cape, J.N. and Unsworth, M.H., 1989. Deposition of atmospheric pollutants on forests. *Philos. Trans. R. Soc. London, Ser. B*, 324: 247-265.
- Gee, A.S., 1990. Introduction to Welsh studies. In: R.W. Edwards, J.H. Stoner and A.S. Gee (Editors), *Acid Waters in Wales*. Kluwer, Dordrecht, pp. 1-9.
- Goenaga, X. and Williams, D.J., 1988. Aluminium speciation in surface waters from a Welsh upland area. *Environ. Pollut.*, 52: 131-150.
- Hauhs, M., 1986. A model of ion transport through a forested catchment at Lange Bramke, West Germany. *Geoderma*, 38: 97-113.
- Hornung, M., Reynolds, B., Stevens, P.A. and Neal, C., 1987. Increased acidity and aluminium concentrations in streams following afforestation. In: *Proc. Int. Symp. Acidification and Water Pathways*, Norwegian National Committee on Hydrology, Bolkesjo, Norway, pp. 259-268.
- Hornung, M., Reynolds, B., Stevens, P.A. and Hughes, S., 1990a. Water quality changes from input to stream. In: R.W. Edwards, J.H. Stoner and A.S. Gee (Editors), *Acid Waters in Wales*. Kluwer, Dordrecht, pp. 223-240.
- Hornung, M., Brown, S.J. and Ranson, A., 1990b. Amelioration of surface water acidity by catchment management. In: R.W. Edwards, J.H. Stoner and A.S. Gee (Editors), *Acid Waters in Wales*. Kluwer, Dordrecht, pp. 311-328.
- Hudson, J., 1988. The contribution of soil moisture storage to the water balances of upland forested and grassland catchments. *Hydrol. Sci. J.*, 33: 289-309.
- Hughes, S. and Reynolds, B., 1988. Cation exchange properties of porous ceramic cups: implications for fieldwork. *Plant Soil*, 109: 141-144.
- Hughes, S. and Reynolds, B., 1990. Evaluation of porous ceramic cups for monitoring soil-water aluminium in acid soils. *J. Soil Sci.*, 41: 325-328.
- Hutson, J.L. and Wagenet, R.J., 1991. Simulating nitrogen dynamics in soils using a deterministic model. *Soil Use Manage.*, 41: 74-78.
- Neal, C., Reynolds, B., Stevens, P.A. and Hornung, M., 1989. Hydrogeochemical controls for

- inorganic aluminium in acidic stream and soil waters at two upland catchments in Wales. *J. Hydrol.*, 106: 155–175.
- Neal, C., Mulder, J., Christophersen, N., Neal, N., Waters, D., Ferrier, R.C., Harriman, R. and McMahon, R., 1990. Limitations to the understanding of ion-exchange and solubility controls for acidic Welsh, Scottish and Norwegian sites. *J. Hydrol.*, 116: 11–23.
- Ormerod, S.J., Weatherley, N.S., Varallo, P.V. and Whitehead, P.G., 1988. Preliminary empirical models of the historical and future impact of acidification on the ecology of Welsh streams. *Freshwater Biol.*, 20: 127–140.
- Ormerod, S.J., Donald, A.P. and Brown, S.J., 1989. The influence of plantation forestry on the pH and aluminium concentrations of upland Welsh streams: a re-examination. *Environ. Pollut.*, 62: 47–62.
- Pyatt, D.G., 1977. Guide to Site Types in Forests of North and Mid-Wales. Forestry Commission Record No. 69. HMSO, London, 51pp.
- Pyatt, D.G. and Smith, K.A., 1983. Water and oxygen regimes of four soil types at Newcastleton Forest, south Scotland. *J. Soil. Sci.*, 34: 465–482.
- Reuss, J.O. and Johnson, D.W., 1986. Acid deposition and the acidification of soils and waters. *Ecol. Stud.* 59: 1–119.
- Reynolds, B., Neal, C., Hornung, M. and Stevens, P.A., 1986. Baseflow buffering of stream-water acidity in five mid-Wales catchments. *J. Hydrol.*, 87: 167–185.
- Reynolds, B., Neal, C., Hornung, M., Hughes, S. and Stevens, P.A., 1988. Impact of afforestation on the soil solution chemistry of stagnopodzols in mid-Wales. *Water, Air Soil Pollut.*, 38: 55–70.
- Seip, H.M., 1980. Acidification of freshwater-sources and mechanisms. In: D. Drablos and A. Tollan (Editors), *Ecological Impact of Acid Precipitation*, Proc. Int. Conf., Sundefjord, Norway, pp. 248–269.
- Soulsby, C., 1991. Hydrological Pathways, Aluminium Mobilization and Streamwater Acidity in an Afforested Catchment in Upland Wales. Ph.D. Thesis, University of Wales (unpublished).
- Soulsby, C., 1992. Acid runoff generation in an afforested headwater catchment at Llyn Brianne, Mid-Wales, *J. Hydrol.* (in press).
- Stevens, P.A., Hornung, M. and Hughes, S., 1989. Solute concentrations, fluxes and major nutrient cycles in a mature sitka spruce plantation in Beddgelert Forest, North Wales. *Forest Ecol. Manage.*, 27: 1–20.
- Stoner, J.H., Gee, A.S. and Wade, K.R., 1984. Effects of acidification on the ecology of streams in the upper Tywi catchment in West Wales. *Environ. Pollut.*, 35: 125–157.
- Talsma, T., 1969. In situ measurement of sorptivity. *Aust. J. Soil Res.*, 7: 269–276.
- United Kingdom Acid Waters Review Group, 1989. *Acidity in United Kingdom Fresh Waters*, HMSO, London, 61pp.
- Van Grinsven, J.J.M., Van Breeman, N. and Mulder, J., 1987. Impacts of acid atmospheric deposition on woodland soils in the Netherlands: 1. Calculation of hydrologic and chemical budgets. *Soil Sci. Soc. Am. J.*, 51: 1629–1634.
- Wagenet, R.J. and Hutson, J.L., 1989. *Leaching Estimation and Chemistry Model (User Manual)*. Water Resources Institute, Cornell University, Vol. 2, pp. 1–148.
- Wheater, H.S., Langan, S.J., Beck, M.B., Jenkins, A., Tuck, S., Miller, J.D. and Ferrier, R.C., 1990. Hydrological processes on the plot and hillslope scales. In: B.J. Mason (Editor), *Surface Water Acidification Programme*, Royal Society, London.

Update

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Erratum

Soulsby, C. and Reynolds, B., 1992. Modelling hydrological processes and aluminium leaching in an acid soil at Llyn Brianne, Mid-Wales. *J. Hydrol.* 138 (1992) 409–429.

p. 416, θ should replace Θ .

Equation (1) should now read:

$$\frac{\delta\theta}{\delta t} = \frac{\delta}{\delta x} [K(\theta) \cdot \delta h / \delta x] - U(x, t)$$

Line 23 should now read:

. . . $K(\theta)$ is the hydraulic conductivity (mm day^{-1}) at water content θ , . . .